BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Inter, Part I, 2023 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. A solution of glucose is 10% w/v. The volume in which 1 g mole of it is dissolved will be:

(A) 200 cm^3
(B) 1 dm^3
(C) 900 cm^3
(D) 1.8 dm^3

Q2. Splitting of spectral lines when atoms are subjected to strong electric field is called:

(A) Stark effect
(B) Zeeman effect
(C) Photoelectric effect
(D) Compton effect

Q3. Pressure remaining constant, at which temperature the volume of a gas will become twice of what it is at 0 degrees C

(A) 546 C
(B) 576K
(C) 200 C
(D) 273K

Q4. Which of the following hydrogen halides has the highest percentage of ionic character:

(A) HF
(B) HBr
(C) HCL
(D) HI

Q5. Calorie is equivalent to

(A) 0.4184 J
(B) 41.84 J
(C) 4.184 J
(D) 418.4 J

Q6. Mass in grams of 2.74 moles of KMnO4:

(A) 0.715 g
(B) 1416.2 g
(C) 432.92 g
(D) 294 g

Q7. The unit of the rate constant is the same as that of the rate of reaction in:

(A) Zero Order Reaction
(B) Second Order
(C) First Order Reaction
(D) Third Order

Q8. Which of the following has hydrogen bonding:

(A) CH4
(B) NaCl
(C) CCL4
(D) NH3

Q9. An excess of aqueous silver nitrate is added to aqueous barium chloride and precipitate is removed by filtration. What are the main ions in the filtrate:

(A) Ag and Ba and NO3, [partially obscured at top edge of scan]
(B) Ag and NO3, only
(C) Ba and NO3 only
(D) Ba and NO3, and [illegible in source scan]

Q10. Isotopes differ in :

(A) Arrangement of electrons
(B) Properties which depend upon mass
(C) Chemical properties
(D) The extent to which they may be affected in electromagnetic field

Q11. The study of heat changes accompanying a chemical reactions is known as:

(A) Physical chemistry
(B) Electro chemistry
(C) Thermochemistry
(D) Analytical chemistry

Q12. Solvent extraction is an equilibrium process and is controlled by:

(A) Law of mass action
(B) The amount of solvent used
(C) Distribution law
(D) The amount of solute

Q13. Stronger the oxidizing agent, greater is the

(A) Oxidation potential
(B) Reduction potential
(C) Redox potential
(D) EMF of cell

Q14. The optimum pressure in ammonia synthesis by Haber's process is:

(A) 100-400 atm
(B) 250-400 atm
(C) 200-300 atm
(D) 150-450 atm

Q15. Cathode in Nickel Cadmium cell is!

(A) Zn
(B) NiO2
(C) Cd
(D) Ag2O

Q16. Ionic solids are characterized by:

(A) Low melting points
(B) High vapour pressures
(C) Good conductivity in solid state
(D) Solubility in polar solvents

Q17. The partial pressure of oxygen in air is:

(A) 160 torr
(B) 159 Torr
(C) 110 torr
(D) 160 torr

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Define percentage yield give example.

ii How many moles of CO2 can be produced from burning one mole of octane mass of octane is 114?

iii Calculate mass in grams of 2.74 mole of KMnO4.

iv How do you differentiate between diffusion and effusion?

v Gases show non ideal behaviour at low temperature and high pressure, give reason.

vi What is Avogadro's law of gases? Give example.

vii Write electronic configuration of C29 and K19

viii Why positive rays are also called as canal rays?

ix Why oxygen molecule is paramagnetic in nature?

x Define state function, give example

xi Justify that heat of formation of a compound is the sum of all the other enthalpies.

xii What is a spontaneous process? Give two examples.

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i What is difference between qualitative and quantitative analysis?

ii Define sublimation. Write name of two compounds which can be sublimed.

iii Write two uses of chromatography.

iv What are intermolecular forces of attraction? Give two examples.

v Evaporation causes cooling. Give reason.

vi Diamond is hard and an electrical insulator. Give reason.

vii Differentiate between hydration and hydrolysis.

viii The concentration in terms of molality is independent of temperature but molarity depends upon temperature. Give reason.

ix Justify that the boiling point of one molal urea solution is 100.52 degrees C but the boiling point of two molal urea solution is less than 101.04 degrees C.

x Define homogeneous catalysis. Give example.

xi Justify that the radioactive decay is always a first order reaction

xii Differentiate between rate and rate constant of a reaction.

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i Define bond order. Give one example

ii Differentiate between bonding molecular orbital and antibonding molecular orbital.

iii The radius of an onion is always larger than parent atom. Why?

iv How does the equilibrium constant of a reaction tell us the direction of a chemical reaction?

v How can NaCl be purified by common ion effect?

vi What are pK, and pK? How do they show the acidic and basic strength?

vii What is the function of salt bridge in galvanic cell?

viii What is anodized aluminium?

ix Calculate the oxidation state of Mn in KMnO, and KMnO

QSECTION-II. Note: Attempt any THREE questions.

QQ.5(a). (Marks 4) Define the following terms with examples: (i) Relative atomic mass (ii) Molecular ion (ii) Isotope (iv) Molar volume

QQ.5(b). (Marks 4) A sample of nitrogen gas is enclosed in a vessel of volume 380 cm3 at 120 degrees C and pressure of 101325 Nm This gas is transferred to a 10 dm3 flask and cooled 27 degrees C Calculate the pressure in Nm exerted by the gas at 27 degrees C.

QQ.6(a). (Marks 4) Describe four properties of the crystalline solids.

QQ.6(b). (Marks 4) What is bomb calorimeter and describe it with the help of diagram?

QQ.7(a). (Marks 4) Derive the equation to calculate radius of electron in nth orbit hydrogen atom by using Bohr's model.

QQ.7(b). (Marks 4) The solubility of PbF2 at 25 degrees C is 0.64 g dm3. Calculate Mass of F-19

QQ.8(a). (Marks 4) Define atomic orbital hybridization. How can we describe the geometry of NH3 on its basis?

QQ.8(b). (Marks 4) What is lead accumulator battery? Discuss its discharging process.

QQ.9(a). (Marks 4) Discuss Raoult's law when one component is volatile other is non-volatile.

QQ.9(b). (Marks 4) Describe half life method and method of large excess for finding the order of reaction.